# Where this stands (11 Nov 2026): a simpler target than the one we sent on 4 Nov

We checked every design against how lithium targets are really made, at LNL and elsewhere. This is what changed. It is a proposal for discussion, and we would like your view

### Nothing in the beam pipe

On 4 Nov we meant a coating chamber. With your pressed lithium not even that is needed.

### Your Cu ogive is close: ×1.0–2.1

That is what the light shells gain on it with today’s detector. The ×2.2–5.2 needs the new calorimeter.

### Li 10–12 µm, not 20

A 20 µm film that turns into LiOH brings the beam down to the 441 keV resonance.

### A flat face at 45°

One piece of aluminium, a Cu stop, your rolled lithium pressed on with flat moulds.

| design | where it stands | why |
|---|---|---|
| **Flat Al end cap at 45°**, Cu stop, your pressed Li | **proposed** | flat parts only, and the lithium is made the way you make it now |
| Your Cu ogive, Li pressed on the tip | **buildable now** | the simplest step up from the cup. We would electroform it in hard copper |
| Al ogive, Cu plated over the whole inside, Li on the tip | **buildable now** | plating Cu on Al has a precedent. No mask, no evaporator |
| Al ogive, Cu at the tip only, Li evaporated through a mask (sent 4 Nov) | **possible, a development project** | three steps we found no precedent for. It gains little with today’s detector |

Every design side by side, with the numbers: [dylan-neff.web.cern.ch/notes/lnl-target-compare.html](dylan-neff.web.cern.ch/notes/lnl-target-compare.html)

# What we learned from how you and others make lithium targets

Lithium metal is always made off the beam line, then carried to it under argon or under vacuum. Nothing needs to be deposited in the beam pipe

| laboratory | how the lithium target is made | source |
|---|---|---|
| **LNL CN, your group** | metal, rolled to ~20 µm (10–40) in an argon glovebox, pressed on a 300 µm Cu cup with paired moulds. air-cooled, ~10 W | arXiv:2605.12005 |
| **LNL AN2000, X17 group** | 24 evaporated LiF films, 34–935 µg/cm², on thin self-supporting backings: 6–8 h of stable rate, the best two days | Góngora-Servín, thesis 2025 |
| **BINP Novosibirsk** | metal, 7–200 µm evaporated on a separate stand. the target leaves under vacuum behind its own gate valve | JINST 15 P10006 |
| **MEG II (PSI)** | LiPON 7 µm sputtered on 25 µm Cu, 8–11 µA for 4 weeks, no significant deterioration | arXiv:2411.07994 |
| **ATOMKI** | LiF, $Li_2O$. metal Li on 10 µm Al diffused into the backing | arXiv:2205.07744 |
| **PTB, Karlsruhe, Notre Dame** | metal, 5–100 µm evaporated on Ta or Cu for neutron production | several papers |

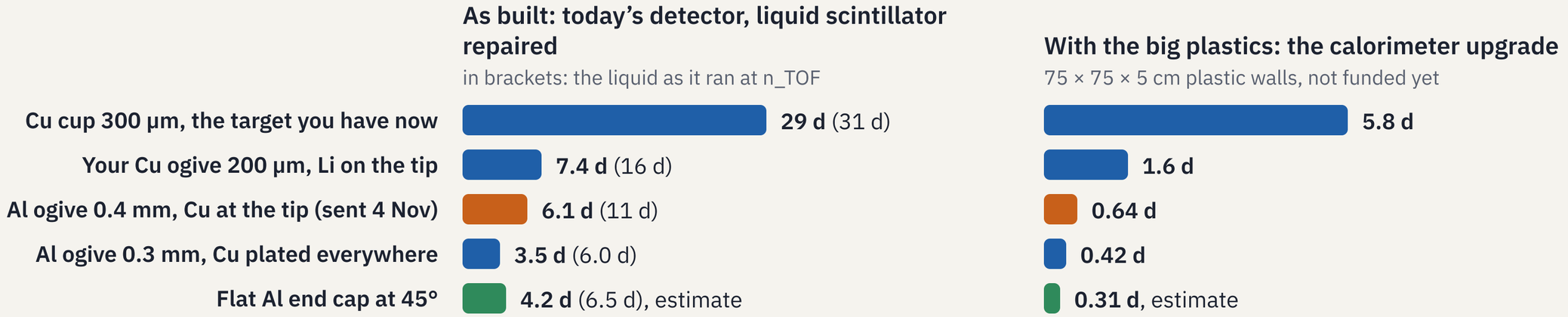
**Your layout is the robust one.** A film on a metal stop that carries the heat lasts. The thin-film route broke at LNL: on AN2000 most LiF films gave 6–8 h of stable rate, the best two days. The one that lasted elsewhere, at MEG II, sat on 25 µm of copper.

**20 µm of rolled lithium pressed on copper is already your routine.** It is exactly our film and our stop, made without any evaporation.

**We do not need anything deposited in the beam pipe.** If our question of last week read that way, it was not what we meant: we had a coating chamber in mind. We found no laboratory that deposits Li, Cu or Ti on the line.

# With today’s detector your Cu ogive is within a factor of 1.0–2.1 of the light shells

Days of beam to 3σ at the ATOMKI strength, at 1 μA and 1.10 MeV, all with 20 μm of Li. The light wall pays once the calorimeter is upgraded

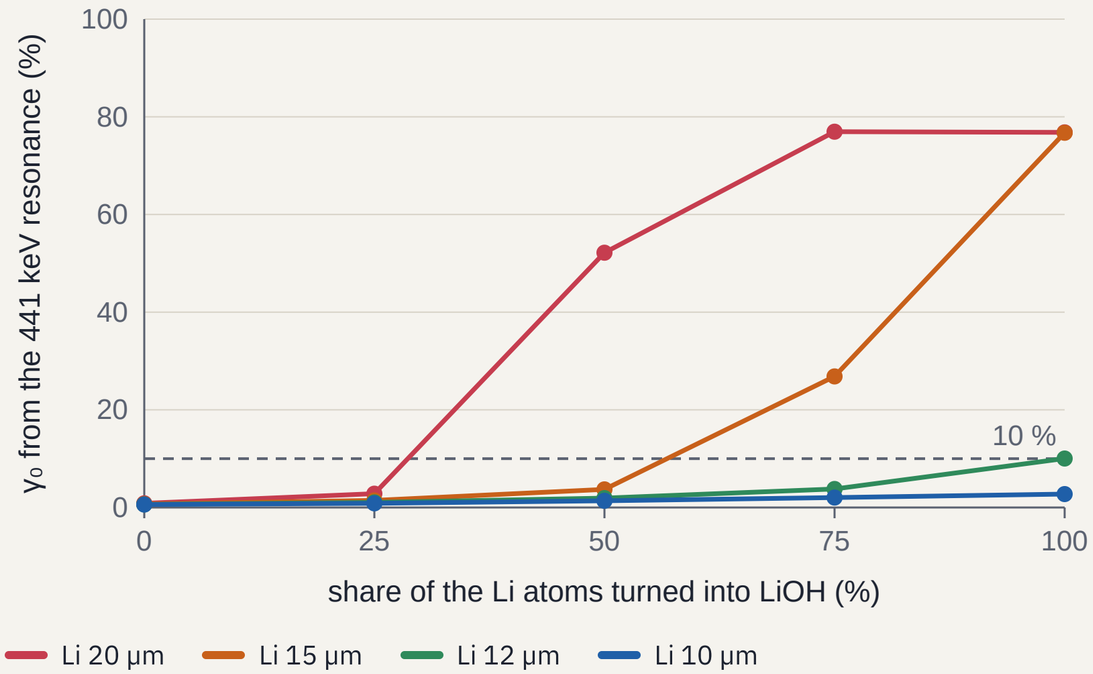



**The big step is yours: from the cup to the ogive, ×3.9 as built.** On top of it the light shells gain ×1.0–2.1 today. The low end is inside what our simulation can resolve, and a larger one is running.

**With the big plastics the light shells gain ×2.2–5.2.** The detector then measures the pair energy well enough for the cleaner opening angle to count. That is when the aluminium wall is worth its trouble.

# The lithium: 10–12 µm instead of 20

Residual water turns Li into LiOH. The lithium atoms stay, but the O and H add stopping, so the beam slows down to the strong 441 keV resonance inside the film. This holds for every target, the cup and the ogive included




Half of a 20 µm Li film turned to LiOH by residual water takes a 1.10 MeV beam down to 454 keV: 52 % of the $\gamma_0$ then comes from the 441 keV resonance (fresh 0.85 %; 91 % at 1.03 MeV).

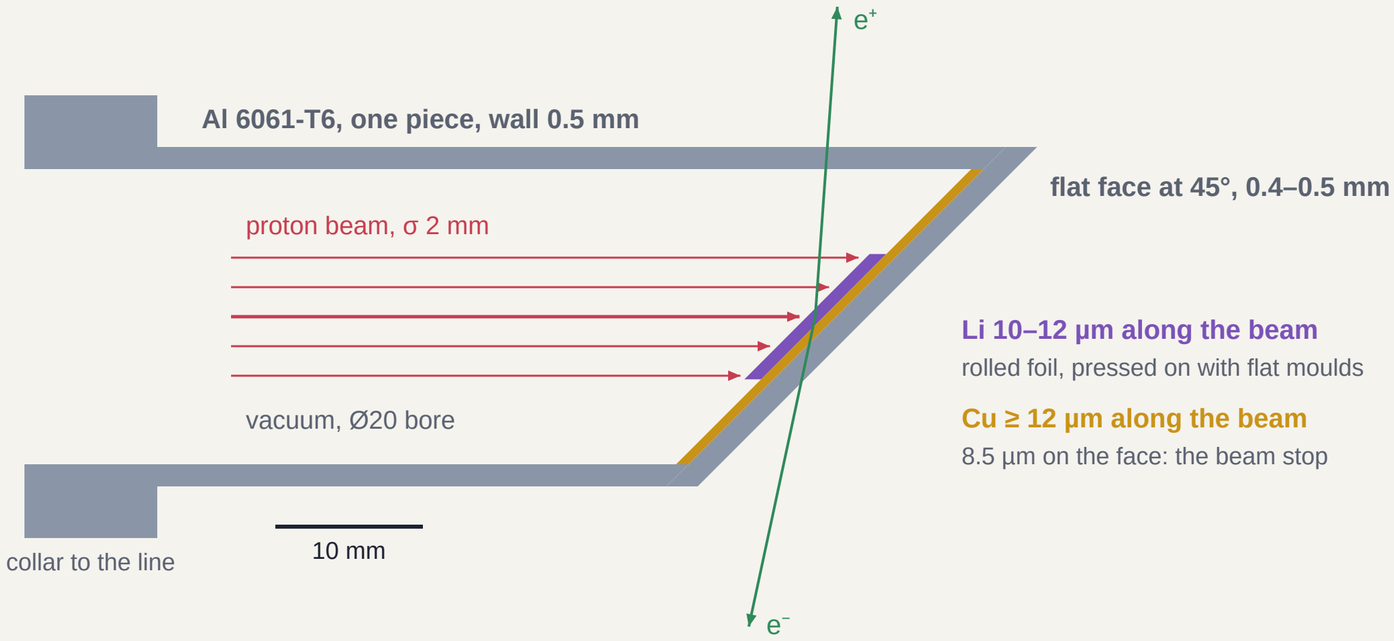
10–12 µm of Li stays at or below 10 % even fully converted, for 51–61 % of the yield. You already roll foils below 10 µm.

| Li along the beam | $\gamma_0$ yield | 441 keV share if all of it is LiOH | days, your Cu ogive | days, flat Al cap (estimate) |
|---|---|---|---|---|
| **20 µm** | 100% | 77% | 1.6 / 7.4 | 0.31 / 4.2 |
| **15 µm** | 77% | 77% | 2.0 / 9.8 | 0.37 / 4.7 |
| **12 µm** | 61% | 10% | 2.4 / 12 | 0.43 / 5.5 |
| **10 µm** | 51% | 2.7% | 2.8 / 14 | 0.51 / 6.3 |

Days to 3σ at 1 µA, 1.10 MeV: big plastics / as built with the liquid repaired. The rate is unknown: 1.35 µm a day at $10^{-7}$ mbar of water if every molecule sticks (an upper limit; LiOH passivates). A one-week film test on the LNL line decides the thickness.

# Proposal: a flat aluminium end cap at 45°, with your pressed lithium

Your window idea at 45°, in plain aluminium: no diamond, no rings, no glue, no Mo. Flat parts only, and the lithium goes on the way you put it on the cup




**Li 10–12 µm along the beam**
rolled foil, pressed on with flat moulds

**Cu ≥ 12 µm along the beam**
8.5 µm on the face: the beam stop

**0.43 d**
to 3σ at 1 µA with the big plastics and 12 µm of Li (20 µm: 0.31 d). Estimate, Geant4 running

**5.5 d**
as built with 12 µm of Li, liquid repaired (as at n_TOF: 9.3 d. 20 µm: 4.2 d). Estimate, Geant4 running

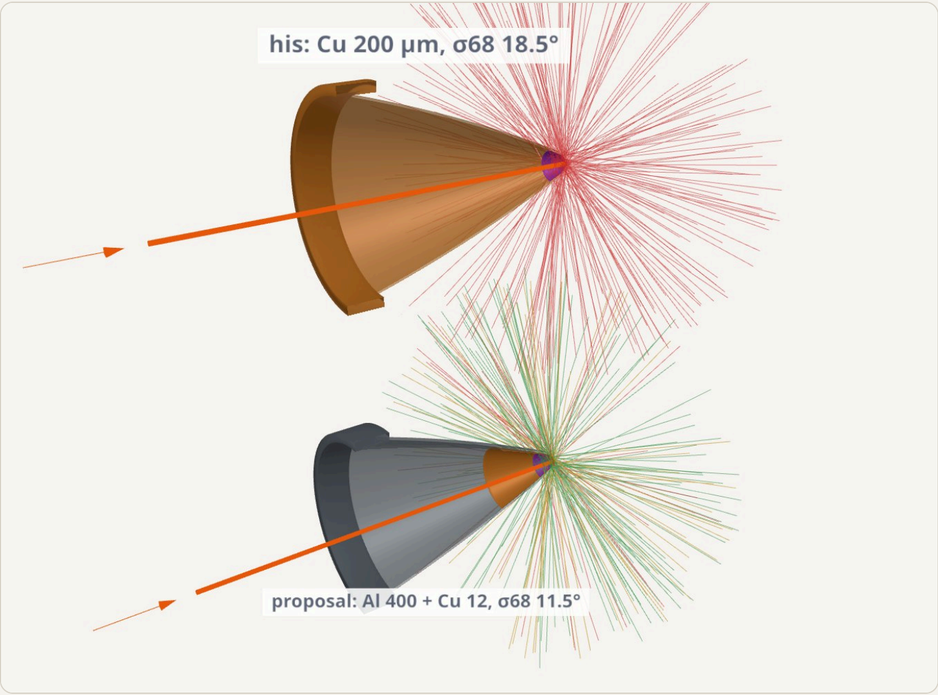
**×3.5–5.4**
margin at 1 bar for a 0.4–0.5 mm face over the Ø20 bore (estimate)

**+5 K**
across a 0.4 mm face at 1 µA, 1.1 W (estimate)

- Flat parts only: a milled face, an ordinary Cu coating, rolled Li pressed with flat moulds as LNL does now.
- The Li is uniform along the beam by geometry: no flank, no mask, no line-of-sight coating.
- No diamond, no rings, no glue, no Mo: Cu K x-rays stop in the 0.4–0.5 mm of aluminium.
- Still open: The Ø20 end piece stays in the leptons' view, as for the diamond window (a 0.3 mm wall is the lever).

# If the ogive stays: two versions that need no new coating tools

Days: big plastics / as built / as built with the liquid as at n_TOF, at 1 µA with 20 µm of Li




Top: your Cu ogive, every lepton crosses 200 µm of Cu (red). Bottom: the Al ogive, legs coloured by the Al they cross.

### 1 · Your Cu ogive, electroformed

**buildable now** · 1.6 / 7.4 / 16 d

Hard copper grown on an aluminium mandrel, 200–250 µm. A spun, annealed 200 µm shell has only ×2–3 on buckling at 1 bar. Li pressed on the tip cap only, so none sits on the steep flank.

### 2 · The Al ogive, 0.4 mm, Cu plated over the whole inside

**buildable now** · 0.42 / 3.5 / 6.0 d (0.3 mm wall simulated)

A thin sputtered seed or a zincate, then electroplated Cu to ≥ 12 µm at the tip, with an anode wire on the axis. Li pressed on the tip. No mask and no evaporator.

### What we sent on 4 Nov: Cu at the tip only, Li evaporated through a mask

**possible, a development project** · 0.64 / 6.1 / 11 d (0.4 mm wall)

It needs a long sputtering run deep in the cone (7–30 h, estimate) and 0.8–5.4 g of lithium per coating. Plating the whole inside loses nothing against it.

## YOUR VIEW?

**1** · How do you intend to put the lithium on your ogive: pressed foil with new moulds for the pointed tip, or evaporation? Your 2015 paper mentions a lithium evaporator. Does it work today, and how does a target leave it?

**2** · Given how your group works, would you rather have a part on which your rolled and pressed lithium works directly?

**3** · Would you consider a **flat face at 45°** instead of the point? It keeps your method and removes the flank.

**4** · Can the LNL workshop turn a **0.4–0.5 mm aluminium wall**? Can the targets group put 10–15 µm of copper on a flat aluminium face below ~150 °C, or would a plating shop do it?

**5** · The **vacuum at the target** on the line you have in mind: pressure, residual gas, PVC or ceramic insulator. Could a film sit there for a week under beam while we watch the 478 keV and $\gamma_0$ lines?

**6** · How thin can a rolled foil be made reproducibly (we would like **8–12 µm**), and how is it carried from the glovebox to the line?

**7** · The **beam spot** at the target, σ and halo, as before.

Every design side by side: dylan-neff.web.cern.ch/notes/lnl-target-compare.html · Interactive 3D: dylan-neff.web.cern.ch/facilities/lnl-target.html